

# Project 1: Real Bussiness Cycles



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**Macroeconomics I**

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## 1. Solve and Log-Linearize

Find equations describing the system. Find the steady state values of control and state parameters for this economy.

### (Household's Problem)

The representative household maximizes its lifetime utility function subject to its budget constraint:

$$\max_{c_t, k_{t+1}, N_t} U = \mathbb{E}_t \left[ \sum_{i=0}^{\infty} \beta^i (\log c_t - A N_t) \right] \quad (1)$$

$$\text{s.t.} \quad c_t + k_{t+1} \leq w_t N_t + (1 + R_t - \delta)k_t + \Pi_t \quad (2)$$

The Lagrangian for household problem is:

$$\mathcal{L} = \mathbb{E}_t \sum_{i=0}^{\infty} \beta^i [(\log c_t - A N_t) - \mu_t (c_t + k_{t+1} - w_t N_t - (1 + R_t - \delta)k_t)] \quad (3)$$

The FOCs are:

$$\frac{\partial \mathcal{L}}{\partial c_t} = \frac{1}{c_t} - \mu_t = 0 \quad (4)$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = A - \mu_t w_t = 0 \quad (5)$$

$$\frac{\partial \mathcal{L}}{\partial k_{t+1}} = \beta \mathbb{E}\{\mu_{t+1}(1 + R_t - \delta)\} - \mu_t = 0 \quad (6)$$

Substituting equation (4) into the equation (5), we get:

$$A c_t = w_t \quad (7)$$

Substituting equation (4) into the equation (6), we get:

$$\beta \mathbb{E}\left\{\frac{1}{c_{t+1}}(1 + R_t - \delta)\right\} = \frac{1}{c_t} \quad (8)$$

**(Firm's Problem)**

The firm's technology is described by the production function:

$$Y_t = z_t k_t^\alpha N_t^{1-\alpha} \quad (9)$$

The technology follows an *AR1* process:

$$\log(z_{t+1}) = \rho \log(z_t) + \epsilon_t, \quad \epsilon_t \sim IID \quad (10)$$

Firm wants to maximize its profit:

$$\max \quad \Pi_t = z_t k_t^\alpha N_t^{1-\alpha} - w_t N_t - R_t k_t \quad (11)$$

The optimality conditions are:

$$MP_L = (1 - \alpha) z_t k_t^\alpha N_t^{-\alpha} = w_t \quad (12)$$

$$MP_K = \alpha z_t k_t^{\alpha-1} N_t^{1-\alpha} = R_t \quad (13)$$

The structural equations of the economy are:

$$\text{(LNS, MARKET CLEARING)} \quad c_t + k_{t+1} = w_t N_t + (1 + R_t - \delta) k_t + \Pi_t \quad (14)$$

$$\text{(PRODUCTION OF ECONOMY)} \quad Y_t = z_t k_t^\alpha N_t^{1-\alpha} \quad (15)$$

### (The Steady State)

In the steady state we have no shock and zero profit ( $\Pi = 0$ ,  $z = 1$ ). Also the sytem parameters has no dynamic. So the equations are:

#### Supply:

$$Ac = w \\ \Rightarrow w = 3c$$

$$\beta \frac{1}{c}(1 + R - \delta) = \frac{1}{c} \\ \Rightarrow R = \frac{1}{\beta} + \delta - 1 \\ \Rightarrow R = \frac{1}{0.99} + 0.025 - 1 = 0.035 \checkmark$$

#### Demand:

$$(1 - \alpha) k^\alpha N^{1-\alpha} = w \\ \Rightarrow (1 - 0.36) k^{0.36} N^{1-0.36} = w \quad (i)$$

$$\alpha k^{\alpha-1} N^{1-\alpha} = R \\ \Rightarrow 0.36 k^{0.36-1} N^{1-0.36} = 0.035 \quad (ii)$$

#### Structural equations:

$$c = wN + (R - \delta)k \\ \frac{w}{3} = wN + (0.035 - 0.025)k \quad (iii)$$

$$Y = k^\alpha N^{1-\alpha}$$

From (i), (ii) we get:

$$\frac{k}{N} = 38, w = 2.37 \quad (iv)$$

From (iii), (iv) we get:

$$k = 10.9, N = 0.28 \quad (\text{v})$$

From (v) and production function, we get:

$$y = 1.06 \quad (\text{vi})$$

From (iv),  $w = 3c$ , we get:

$$c = 0.79 \quad (\text{vii})$$

From (v) and law of motion of capital,  $0 = k(-\delta) + i$ , we get:

$$i = 0.27$$

## 2. Log-Linearization

**Supply:**

$$c_t = Aw_t \quad (1)$$

$$\beta \mathbb{E} \left\{ \frac{1}{c_{t+1}} (1 + R_t - \delta) \right\} = \frac{1}{c_t} \quad (2)$$

**Demand:**

$$(1 - \alpha) z_t k_t^\alpha N_t^{-\alpha} = w_t \quad (3)$$

$$\alpha z_t k_t^{\alpha-1} N_t^{1-\alpha} = R_t \quad (4)$$

**Structural equations:**

$$Y_t = c_t + i_t \quad (5)$$

$$Y_t = z_t k_t^\alpha N_t^{1-\alpha} \quad (6)$$

$$\log(z_{t+1}) = \rho \log(z_t) + \epsilon_t \quad (7)$$

$$k_{t+1} = k_t(1 - \delta) + i_t \quad (8)$$

Log-linearizing (1), we get:

$$\tilde{c}_t = \tilde{w}_t \quad (i)$$

Log-linearizing (2), we get:

$$\mathbb{E}(\tilde{c}_{t+1}) = \beta \mathbb{E}(\tilde{R}_{t+1}) + \tilde{c}_t \quad (ii)$$

Log-linearizing (3), we get:

$$\tilde{z}_t + \alpha \tilde{k}_t - \alpha \tilde{N}_t = \tilde{w}_t \quad (iii)$$

Log-linearizing (4), we get:

$$\tilde{z}_t + (\alpha - 1) \tilde{k}_t - (\alpha - 1) \tilde{N}_t = \mathbb{E}(\tilde{R}_{t+1}) \quad (iv)$$

Log-linearizing (5), we get:

$$\tilde{Y} = \frac{c}{Y} \tilde{c}_t + \frac{i}{Y} \tilde{i}_t \quad (v)$$

Log-linearizing (6), we get:

$$\tilde{Y}_t = \alpha \tilde{k}_t + (1 - \alpha) \tilde{N}_t \quad (\text{vi})$$

Log-linearizing (7), we get:

$$\tilde{z}_{t+1} = \rho \tilde{z}_t \quad (\text{vii})$$

Log-linearizing (8), we get:

$$\tilde{k}_{t+1} = (1 - \delta) \tilde{k}_t + \delta \tilde{i}_t \quad (\text{viii})$$

## 2. Linear Rational Expectation Model

$$\tilde{z}_t = \frac{\tilde{z}_{t+1}}{\rho} \quad (\text{i})$$

$$\tilde{c}_t = \beta \mathbb{E} \left( \tilde{R}_{t+1} \right) - \mathbb{E} \left( \tilde{c}_{t+1} \right) \quad (\text{ii})$$

$$\tilde{w}_t = \tilde{c}_t = \beta \mathbb{E} \left( \tilde{R}_{t+1} \right) - \mathbb{E} \left( \tilde{c}_{t+1} \right) \quad (\text{iii})$$

$$\left\{ \begin{array}{l} \tilde{Y}_t = \frac{c}{Y} \beta \mathbb{E} \left( \tilde{R}_{t+1} \right) - \mathbb{E} \left( \tilde{c}_{t+1} \right) + \frac{i}{Y} \left[ \frac{1}{\delta} \tilde{k}_{t+1} - \frac{1-\delta}{\delta} \tilde{k}_t \right] \\ \tilde{Y}_t = \alpha (\tilde{k}_t - \tilde{N}_t) + \tilde{N}_t \\ \alpha (\tilde{k}_t - \tilde{N}_t) = \beta \mathbb{E} \left( \tilde{R}_{t+1} \right) - \mathbb{E} \left( \tilde{c}_{t+1} \right) - \frac{\tilde{z}_{t+1}}{\rho} \end{array} \right\} \implies \tilde{Y}_t = \beta \mathbb{E} \left( \tilde{R}_{t+1} \right) - \mathbb{E} \left( \tilde{c}_{t+1} \right) - \frac{\tilde{z}_{t+1}}{\rho} + \tilde{N}_t \quad \left. \vphantom{\left\{ \right.} \right\}$$

$$\implies \tilde{N}_t + \frac{i}{Y} \left[ \frac{\delta-1}{\delta} \right] \tilde{k}_t = \left( \frac{c}{Y} - 1 \right) \beta \mathbb{E} \left( \tilde{R}_{t+1} \right) + \frac{i}{Y} \left[ \frac{1}{\delta} \tilde{k}_{t+1} \right] + \frac{\tilde{z}_{t+1}}{\rho} \quad \left. \vphantom{\left\{ \right.} \right\}$$

$$\tilde{z}_t + (\alpha - 1) \tilde{k}_t - (\alpha - 1) \tilde{N}_t = \mathbb{E} \left( \tilde{R}_{t+1} \right) \quad \left. \vphantom{\left\{ \right.} \right\}$$

$$\tilde{c}_t = \tilde{w}_t \quad (\text{i})$$

$$\mathbb{E} \left( \tilde{c}_{t+1} \right) = 0.99 \mathbb{E} \left( \tilde{R}_{t+1} \right) + \tilde{c}_t \quad (\text{ii})$$

$$\tilde{z}_t + 0.36 \tilde{k}_t - 0.36 \tilde{N}_t = \tilde{w}_t \quad (\text{iii})$$

$$\tilde{z}_t + (0.36 - 1) \tilde{k}_t - (0.36 - 1) \tilde{N}_t = \mathbb{E} \left( \tilde{R}_{t+1} \right) \quad (\text{iv})$$

$$\tilde{Y}_t = \frac{0.79}{1.06} \tilde{c}_t + \frac{0.27}{1.06} \tilde{i}_t \quad (\text{v})$$

$$\tilde{Y}_t = 0.36 \tilde{k}_t + (1 - 0.36) \tilde{N}_t \quad (\text{vi})$$

$$\tilde{z}_{t+1} = 0.95 \tilde{z}_t \quad (\text{vii})$$

$$\mathbb{E} \left( \tilde{k}_{t+1} \right) - (1 - 0.025) \tilde{k}_t = 0.025 \tilde{i}_t \quad (\text{viii})$$

$$\begin{aligned}
 (i, ii, iii, vii) &\implies 0.36\tilde{k}_t - 0.36\tilde{N}_t = \mathbb{E}(\tilde{c}_{t+1}) - 0.99\mathbb{E}(\tilde{R}_{t+1}) - \frac{\mathbb{E}(\tilde{z}_{t+1})}{0.95} \\
 (ii, v, vi) &\implies \frac{0.79}{1.06}(\mathbb{E}(\tilde{c}_{t+1}) - 0.99\mathbb{E}(\tilde{R}_{t+1})) + \frac{0.27\mathbb{E}(\tilde{k}_{t+1}) - (1 - 0.025)\tilde{k}_t}{0.025} = 0.36\tilde{k}_t + (0.64)\tilde{N}_t \\
 (vii) &\implies \tilde{z}_{t+1} = 0.95\tilde{z}_t \\
 (ii) &\implies \mathbb{E}(\tilde{c}_{t+1}) = 0.99\mathbb{E}(\tilde{R}_{t+1}) + \tilde{c}_t
 \end{aligned}$$

Solving the system with Python ([Click Here to Open](#)), we get:

$$\tilde{k}_t = -0.228\mathbb{E}(R_{t+1}) + 0.231\mathbb{E}(c_{t+1}) + 0.932\mathbb{E}(k_{t+1}) - 0.171\mathbb{E}(z_{t+1}) \quad (\text{ix})$$

$$\tilde{N}_t = 2.522\mathbb{E}(R_{t+1}) - 2.547\mathbb{E}(c_{t+1}) + 0.932\mathbb{E}(k_{t+1}) + 2.753\mathbb{E}(z_{t+1}) \quad (\text{x})$$

$$\tilde{z}_t = 1.053\mathbb{E}(\tilde{z}_{t+1}) \quad (\text{xi})$$

$$\tilde{c}_t = -0.990\mathbb{E}(\tilde{R}_{t+1}) + \mathbb{E}(\tilde{c}_{t+1}) \quad (\text{xii})$$

Let:

$$\tilde{X}_t = \begin{bmatrix} \tilde{k}_t \\ \tilde{N}_t \\ \tilde{z}_t \\ \tilde{c}_t \end{bmatrix}$$

Then, we can rewrite the system as:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \tilde{k}_t \\ \tilde{N}_t \\ \tilde{z}_t \\ \tilde{c}_t \end{bmatrix} = \begin{bmatrix} -0.228 & 0.231 & 0.932 & -0.171 \\ 2.522 & -2.547 & 0.932 & 2.753 \\ 0 & 0 & 1.053 & 0 \\ -0.990 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbb{E}(\tilde{k}_{t+1}) \\ \mathbb{E}(\tilde{N}_{t+1}) \\ \mathbb{E}(\tilde{z}_{t+1}) \\ \mathbb{E}(\tilde{c}_{t+1}) \end{bmatrix}$$

Multiplying both sides by  $I_4^{-1}$  from left:

$$\begin{bmatrix} \tilde{k}_t \\ \tilde{N}_t \\ \tilde{z}_t \\ \tilde{c}_t \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} -0.228 & 0.231 & 0.932 & -0.171 \\ 2.522 & -2.547 & 0.932 & 2.753 \\ 0 & 0 & 1.053 & 0 \\ -0.990 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbb{E}(\tilde{k}_{t+1}) \\ \mathbb{E}(\tilde{N}_{t+1}) \\ \mathbb{E}(\tilde{z}_{t+1}) \\ \mathbb{E}(\tilde{c}_{t+1}) \end{bmatrix}$$

Simplifying, we get:

$$\tilde{X}_t = \begin{bmatrix} -0.228 & 0.231 & 0.932 & -0.171 \\ 2.522 & -2.547 & 0.932 & 2.753 \\ 0 & 0 & 1.053 & 0 \\ -0.990 & 1 & 0 & 1 \end{bmatrix} \mathbb{E}(\tilde{\mathbf{X}}_{t+1})$$

So  $A^{-1}B$  is:

$$A^{-1}B = \begin{bmatrix} -0.228 & 0.231 & 0.932 & -0.171 \\ 2.522 & -2.547 & 0.932 & 2.753 \\ 0 & 0 & 1.053 & 0 \\ -0.990 & 1 & 0 & 1 \end{bmatrix}$$

### 3. Eigen-values

Running the MATLAB code ([Click Here to Open](#)), we will reach to the following eigenvalues:

$$\lambda_1 = -3.4346 \quad (1)$$

$$\lambda_2 = 0.0007 \quad (2)$$

$$\lambda_3 = 1.6590 \quad (3)$$

$$\lambda_4 = -1.0530 \quad (4)$$

The only eigenvalue satisfying  $|\lambda| < 1$  is  $\lambda_2 = 0.00067497$ , which is corresponding with:

$$\vec{v}_2 = \begin{bmatrix} 0.71066 & 0.70353 & 0 & 2.6584e-05 \end{bmatrix}$$

## 4. Dynare

To open the Dynare script for this problem: [Click Here](#)

Here are  $\LaTeX$  output of the Dynare model.

| Table 1: Endogenous Variables |          |                   |
|-------------------------------|----------|-------------------|
| Variable                      | $\LaTeX$ | Description       |
| Y                             | $Y$      | Output            |
| I                             | $I$      | Investment        |
| K                             | $K$      | Capital           |
| N                             | $N$      | Labor             |
| A                             | $A$      | Technology        |
| C                             | $C$      | Consumption       |
| w                             | $w$      | Wages             |
| Rk                            | $R_k$    | Return on Capital |

| Table 2: Exogenous Variables |          |                            |
|------------------------------|----------|----------------------------|
| Variable                     | $\LaTeX$ | Description                |
| eA                           | $e_A$    | Exogenous Technology Shock |

| Table 3: Parameters |          |                                 |
|---------------------|----------|---------------------------------|
| Variable            | $\LaTeX$ | Description                     |
| beta                | $\beta$  | Discount Factor                 |
| delta               | $\delta$ | Depreciation Rate               |
| alpha               | $\alpha$ | Output Elasticity of Capital    |
| rhoA                | $\rho_A$ | Persistence of Technology Shock |
| con                 | $con$    | Leisure Utility Constant        |

Static Content:

$$\frac{1}{C} = \frac{1}{C} \beta (Rk + 1 - \delta) \quad (1)$$

$$w = C A \quad (2)$$

$$w = A (1 - \alpha) K^\alpha N^{(-\alpha)} \quad (3)$$

$$R_k = A \alpha K^{\alpha-1} N^{1-\alpha} \quad (4)$$

$$Y = N^{1-\alpha} A K^\alpha \quad (5)$$

$$Y = C + I \quad (6)$$

$$K = I + (1 - \delta) K \quad (7)$$

$$\log(A) = \log(A) \rho_A + eA \quad (8)$$

Dynamic Content:

$$\frac{1}{C_t} = \beta \frac{1}{C_{t+1}} (R_{kt+1} + 1 - \delta) \quad (9)$$

$$w_t = C_t A_t \quad (10)$$

$$w_t = A_t (1 - \alpha) K_t^\alpha N_t^{(-\alpha)} \quad (11)$$

$$R_{kt+1} = A_t \alpha K_t^{\alpha-1} N_t^{1-\alpha} \quad (12)$$

$$Y_t = N_t^{1-\alpha} A_t K_t^\alpha \quad (13)$$

$$Y_t = C_t + I_t \quad (14)$$

$$K_{t+1} = I_t + (1 - \delta) K_t \quad (15)$$

$$\log(A_{t+1}) = \rho_A \log(A_t) + eA_t \quad (16)$$

Original Content:

$$\frac{1}{C_t} = \beta \frac{1}{C_{t+1}} (R_{k_{t+1}} + 1 - \delta) \quad (17)$$

$$w_t = C_t A_t \quad (18)$$

$$w_t = A_t (1 - \alpha) K_t^\alpha N_t^{(1-\alpha)} \quad (19)$$

$$R_{k_{t+1}} = A_t \alpha K_t^{\alpha-1} N_t^{1-\alpha} \quad (20)$$

$$Y_t = N_t^{1-\alpha} A_t K_t^\alpha \quad (21)$$

$$Y_t = C_t + I_t \quad (22)$$

$$K_{t+1} = I_t + (1 - \delta) K_t \quad (23)$$

$$\log(A_{t+1}) = \rho_A \log(A_t) + e A_t \quad (24)$$

STEADY-STATE RESULTS:

$$Y = 3.18801$$

$$I = 0.817411$$

$$K = 32.6964$$

$$N = 0.86068$$

$$A = 1$$

$$C = 2.37059$$

$$w = 2.37059$$

$$R_k = 0.035101$$

$$\rho_A = 3$$

**Stochastic Model Simulation IRF:**

Figure 1: Investment Response

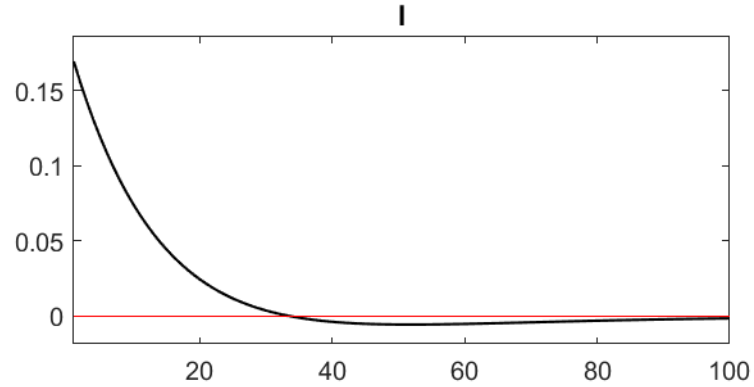


Figure 2: Consumption Response

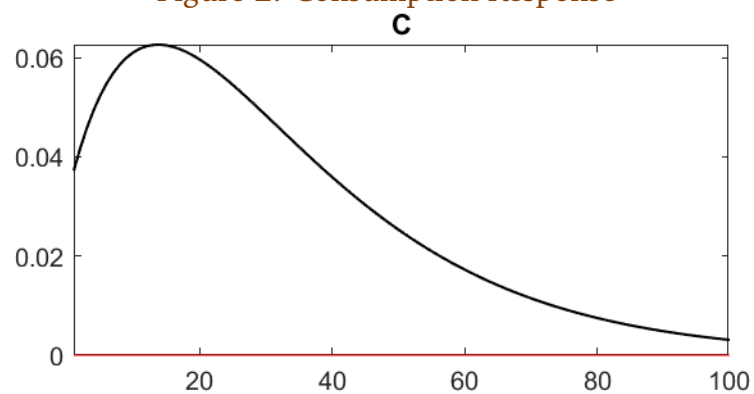


Figure 3: Labor Response

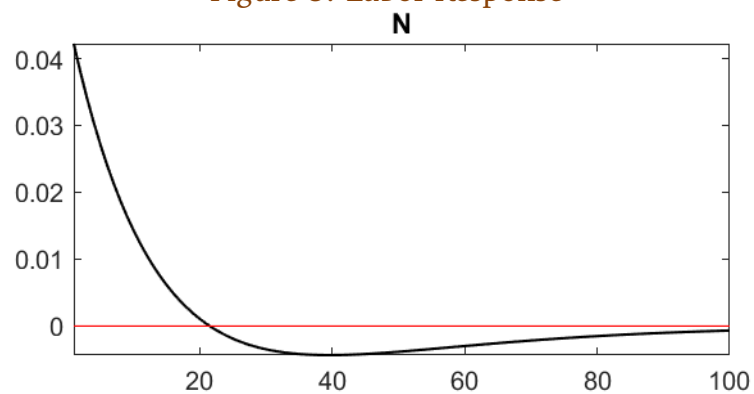


Figure 4: Capital Response

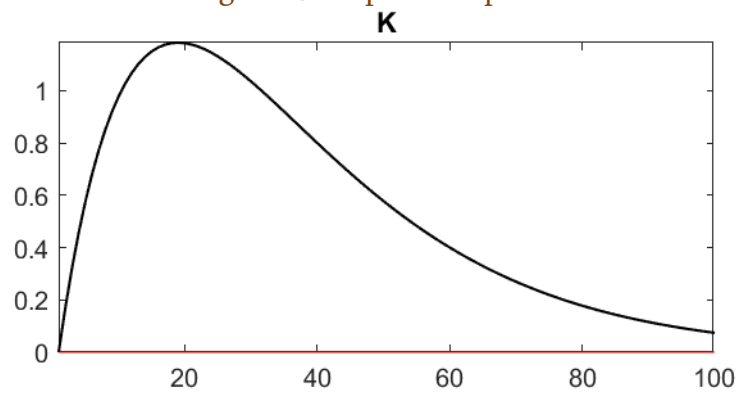


Figure 5: Interest Rate Response

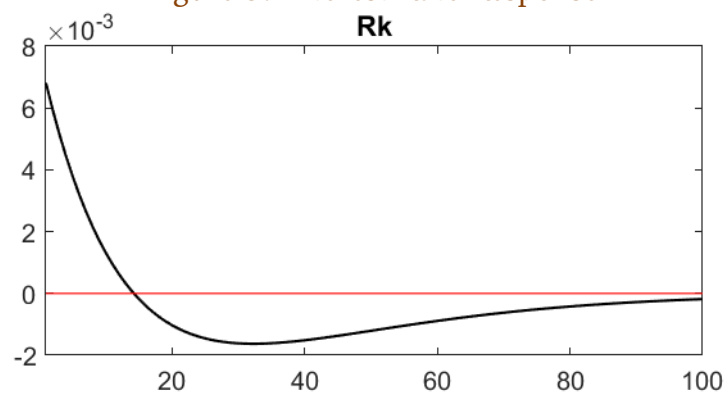


Figure 6: Wage Response

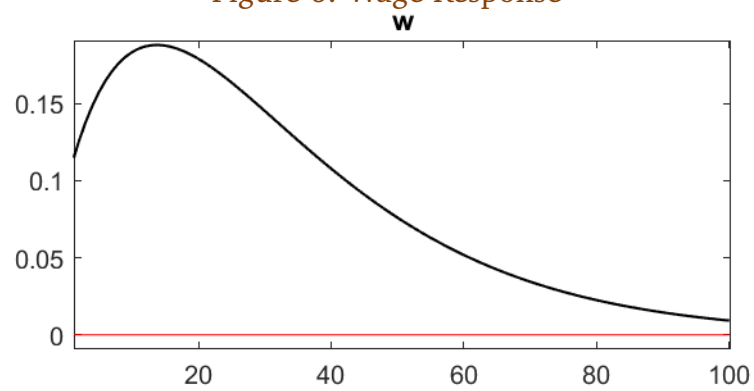


Figure 7: Technology Response

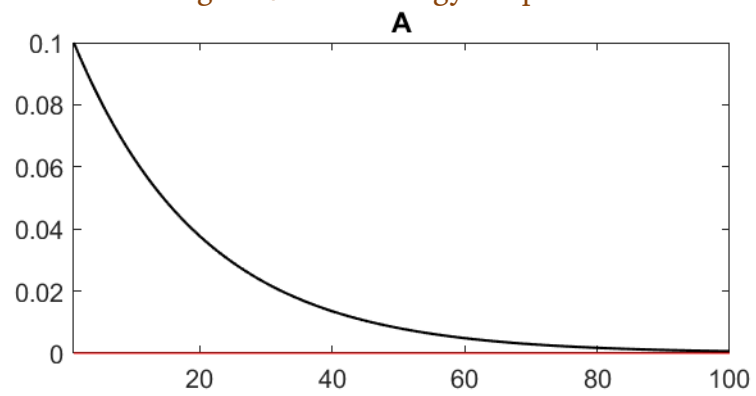
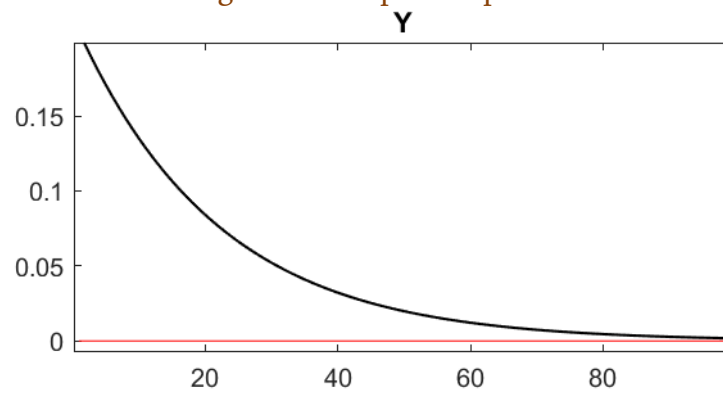


Figure 8: Output Response



## 5. Interpret

Productivity,  $\tilde{z}_t$ , increases, leading to an immediate rise in the marginal product of labor and capital.

Wages  $\tilde{w}_t$  and the rental rate on capital  $\tilde{R}_{kt}$  increase, reflecting the higher productivity.

Consumption,  $\tilde{c}_t$ , jumps due to the income effect of higher wages.

Investment,  $\tilde{i}_t$ , increases as firms seek to capitalize on the higher productivity by expanding their capital stock.

Labor supply is affected by two opposite effects; the substitution effect (working more due to higher wages) and the income effect (working less due to higher consumption).

Labor demand increases due to higher productivity, leading to an increase in employment,  $\tilde{N}_t$  and wages  $\tilde{N}_t$ .

The capital stock  $\tilde{k}_t$  grows over time as investment continues, leading to an expanded productive capacity of the economy.

Output  $\tilde{Y}_t$  increases immediately due to higher productivity.

Over time, as the shock fades out, the output initially reaches its peak and then begins to decrease. This decrease in output leads to a decrease in demand for labor and capital, resulting in lower wages and interest rates. This, in turn, affects the supply side, leading to a further decrease in output, creating a vicious cycle of continued decline.